LESE152: Statistics for Economists

Problem Set 1

Name: _____

ID: _____

Useful values ($\Phi(z) = \Pr(Z \le z)$, $Z \sim N(0,1)$): $\Phi(0.5) = 0.6915$, $\Phi(1) = 0.8413$, $\Phi(1.2816) = 0.90$, $\Phi(1.5) = 0.9332$, $\Phi(2) = 0.9772$, $\Phi(1.645) = 0.95$, $\Phi(1.96) = 0.975$, $e^{-2} = 0.1353$. By symmetry $\Phi(-z) = 1 - \Phi(z)$.

Part A: Multiple Choice Questions (2 points each)

MC1. Which of the following is a *nominal* variable?

(a) Household income (won)

(b) Province of residence

(c) Customer satisfaction rating (1 = very poor, ..., 5 = excellent)

(d) Calendar year

MC2. According to Chebyshev's inequality, for *any* data set, at least what fraction of observations lie within 3 standard deviations of the mean?

(a) 68%

(b) 75%

(c) 88.9%

(d) 99.7%

MC3. Events A and B are independent with $\Pr(A) = 0.6$ and $\Pr(B) = 0.3$. What is $\Pr(A \cup B)$?

(a) 0.18

(b) 0.72

(c) 0.82

(d) 0.90

MC4. A grain elevator receives on average 2 trucks per hour, and the number of arrivals follows a Poisson distribution. What is the probability that no truck arrives in a given hour?

(a) 0

(b) 0.135

(c) 0.271

(d) 0.5

MC5. A delivery arrives at a time uniformly distributed between 20 and 50 minutes after ordering. What is the probability that it takes more than 40 minutes?

(a) 1/4
(b) 1/3
(c) 1/2
(d) 2/3

MC6. Korea's annual rice production recorded for each year from 1990 to 2025 is an example of:

(a) Time-series data
(b) Cross-sectional data
(c) Panel data
(d) Experimental data

MC7. A data set has $Q_1 = 20$ and $Q_3 = 30$. Using the $1.5 \times IQR$ rule, observations above which value are flagged as potential outliers?

(a) 35
(b) 40
(c) 45
(d) 50

MC8. A survey of 500 adults records education and employment status:

	Employed	Not employed	Total
College degree	180	20	200
No college degree	240	60	300
Total	420	80	500

What is Pr(college degree | employed)?

(a) 0.36
(b) 0.43
(c) 0.60
(d) 0.90

MC9. Each of 20 seedlings independently dies with probability 0.1. What is the variance of the number of seedlings that die?

(a) 0.18
(b) 1.8
(c) 2
(d) 18

MC10. Revenue X and cost Y are independent, with standard deviations 6 and 8 (million won). What is the standard deviation of profit $X - Y$?

(a) 2

(b) 10

(c) 14

(d) 100

Part B: True/False Questions (3 points each, Provide Brief Explanation)

TF1. Statement: Postal codes are quantitative variables because they are recorded as numbers, so their mean is a meaningful summary.

- True ☐ False ☐
- Explanation:

TF2. Statement: In the frequentist view of probability, an unknown parameter such as the population mean μ is treated as a random variable with its own probability distribution.

- True ☐ False ☐
- Explanation:

TF3. Statement: If 10 is added to every observation in a data set, the mean increases by 10 but the standard deviation is unchanged.

- True ☐ False ☐
- Explanation:

TF4. Statement: For any two events with positive probability, $\Pr(A \mid B) = \Pr(B \mid A)$.

- True ☐ False ☐
- Explanation:

TF5. Statement: For a continuous random variable, the density $f(x)$ is a probability, so it can never exceed 1.

- True ☐ False ☐
- Explanation:

TF6. Statement: Because a census covers the entire population, its results are free of error.

- True ☐ False ☐
- Explanation:

TF7. Statement: The coefficient of variation can be used to compare the variability of two variables measured in different units (e.g., rice price in won/kg and yield in t/ha).

- True ☐ False ☐
- Explanation:

TF8. Statement: If $\Pr(A) = 0.3$ and $\Pr(B) = 0.5$, then $\Pr(A \cap B) = 0.15$.

- True ☐ False ☐
- Explanation:

TF9. Statement: For a Poisson random variable, the mean and the variance are equal.

- True ☐ False ☐
- Explanation:

TF10. Statement: If a machine's lifetime is exponentially distributed, a machine that has already run 100 hours has the same probability of lasting another 50 hours as a brand-new machine.

- True ☐ False ☐
- Explanation:

Part C: Short Answers

C1. Daily visitors to a farmers' market over 8 days:

$$30, \quad 32, \quad 35, \quad 38, \quad 40, \quad 42, \quad 45, \quad 78$$

(a) Calculate the mean and the median. What does their comparison tell you about the shape of the distribution? (5 pt)

(b) Find Q_1 and Q_3 using the position rule $L_p = \frac{p}{100}(n+1)$ with linear interpolation. Use the $1.5 \times IQR$ rule to check whether any observation is a potential outlier. (5 pt)

(c) Calculate the sample standard deviation s and the coefficient of variation. (5 pt)

C2. Pheromone traps are used to detect a crop pest. 10% of fields are infested. In an infested field, a trap signals with probability 0.80; in a clean field, it (falsely) signals with probability 0.15.

(a) What is the probability that a trap in a randomly chosen field signals? (5 pt)

(b) A trap signals. What is the probability that the field is actually infested? (5 pt)

(c) A second, independent trap is placed in the same field and it also signals. What is the probability that the field is infested given both traps signal? (5 pt)

C3. A rice-cake shop receives X catering orders per day, with the following probability mass function:

x	0	1	2	3
$\Pr(X = x)$	0.2	0.4	0.3	0.1

(a) Calculate $E(X)$ and the standard deviation of X. (5 pt)

(b) Each order earns 40 (thousand won) and the shop pays a fixed daily cost of 10, so daily profit is $Y = 40X - 10$. Calculate $E(Y)$ and $\mathrm{sd}(Y)$. (5 pt)

(c) On a day with 3 orders, each customer pays late independently with probability 0.2. What is the probability that exactly one customer pays late? (5 pt)

(d) Preparation time for an order is $T \sim N(45, 6^2)$ minutes. What is the probability that preparation takes longer than 57 minutes? What preparation time is not exceeded on 90% of orders? (5 pt)